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1. Assume that the individual stages of a datapath have the following latencies:

IF	ID	EX	MEM	WB
200ps	300ps	300ps	250ps	270ps

The following instruction set is executed sequentially (assume no data hazard):

add x5, x1, x2
lw x6, 0(x5)
sw x6, 4(x1)

- Determine the clock cycle time for:
 - Single-cycle datapath
 - Pipelined datapath
- Determine the total number of cycles required for:
 - Single-cycle datapath
 - Pipelined datapath
- Compute the total execution time for this instruction set:
 - Single-cycle datapath
 - Pipelined datapath
- Which datapath design runs this faster and by how much?

Sol:

- add: $200+300+300+270=1070\text{ps}$, lw: $200+300+300+250+270=1320\text{ps}$, sw = $200+300+300+250=1050\text{ps}$

Single cycle = 1320ps

ii. Pipeline = 300ps

- 3 cycles
 - 7 cycles
- $1320 \times 3 = 3960\text{ps}$
 - $300 \times 7 = 2100\text{ps}$
- Pipeline is faster by $(3960-2100) = 1860\text{ps}$

2. Solve the data hazards for the given sequence of RISC-V assembly instructions using Stalling and Forwarding. [7]

1. lw x5, 0(x1)
2. add x6, x5, x2
3. srli x7, x6, 2
4. and x8, x7, x4
5. sw x8, 0(x2)

Clock Cycles	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12				
\	IF	ID	EX	MEM	WB											
		X	X	X	X	X										
			IF	ID	EX	MEM	WB									
				IF	ID	EX	MEM	WB								
					IF	ID	EX	MEM	WB							
						X	X	X	X	X						
							X	X	X	X	X					
								IF	ID	EX	MEM	WB				

Find the Average CPI. [1]

$$\text{AVG CPI} = 12/5 = 2.4$$